

Alumni Subsystem Database Design

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Executive Summary:

The University Alumni Subsystem is a relational database designed to centralize and manage records pertaining to former university students. The subsystem serves four core functions: generating mailing lists for university communications, maintaining accurate and current alumni records, tracking alumni employment outcomes, and supplying data listings to academic departments and administrative units.

The database encompasses six primary entities: Alumni Information, Current Employment, Professional Affiliations, Undergraduate Affiliations, Alumni Association Membership, and Contributions structured to enforce referential integrity and minimize data redundancy. The subsystem is also expected to interact closely with the University Transcript Subsystem, as academic history data will need to be referenced from existing student records.

Key challenges include the dynamic nature of alumni data, the need to scale with a growing alumni population each semester, and the protection of sensitive personal and financial information through strict access controls. To ensure all requirements are met, interviews will be conducted with relevant university staff. Final deliverables will include an Entity-Relationship Diagram, relational schema, data dictionary, and SQL implementation code.

Problem Statement:

Howard University needs a system that can reliably store and manage alumni information as students join the university's alumni community. Without a central structure, important data such as contact details, alumni's current employment, organizational involvement, and contribution history gets scattered across departments. This makes it harder for the university to communicate, support reporting, and keep alumni engaged over time.

To address these problems, the Alumni Subsystem would bring together all key post-graduation data using a relational framework organized into six main areas: Alumni Information, Contributions, Current Employment, Undergraduate Affiliation, Professional Affiliation, and Alumni Association Membership. This setup should maintain data relationships, ensure accuracy as records change, and provide a reliable base for administrative tasks that require up-to-date, accurate alumni information.

The system must also be designed with growth, interoperability, and security in mind. Alumni, the system should also be built to handle growth, work well with other university systems, and keep data secure. As alumni records grow, the subsystem needs to scale while remaining compatible with current systems, including those used for academic verification. Since the database contains sensitive personal and financial information, it is important to have strong controls over who can access and modify the data.

Project Objective:

The objective of this project is to design and build a centralized alumni database that reliably stores, organizes, and manages all important alumni information after graduation. This database will help with communication, reporting, and long-term needs for the institution. By focusing on six main entities, the system will improve data accuracy, reduce redundancy, and provide a secure, scalable solution.

Problems and Constraints:

The Alumni subsystem has some challenges. It needs frequent updates to keep information accurate and to provide complete, accurate personal information. Alumni details, such as addresses, phone numbers, jobs, and contributions, often change. Because it stores sensitive information, the system must follow strict privacy rules. It needs to manage more alumni records each semester, so it should be able to scale and remain reliable as the database grows. Furthermore, the subsystem should be flexible enough to meet the needs of different departments.

System Analysis:

The system requires the ability to record detailed alumni demographics, including names, graduation years, and contact information. It must also track employment history, organizational affiliations, and any contributions made by alumni. Additionally, the system should support efficient searching, updating, and exporting of alumni records, as well as generating mailing lists and formatted reports for departmental use. To ensure quality and usability, the system must maintain accurate, up-to-date data through validation checks, be scalable to handle a large number of records, and provide a user-friendly interface that staff can operate with minimal or no training.

Business Rules:

Each alumnus can make many contributions. Each contribution is made by one alum.

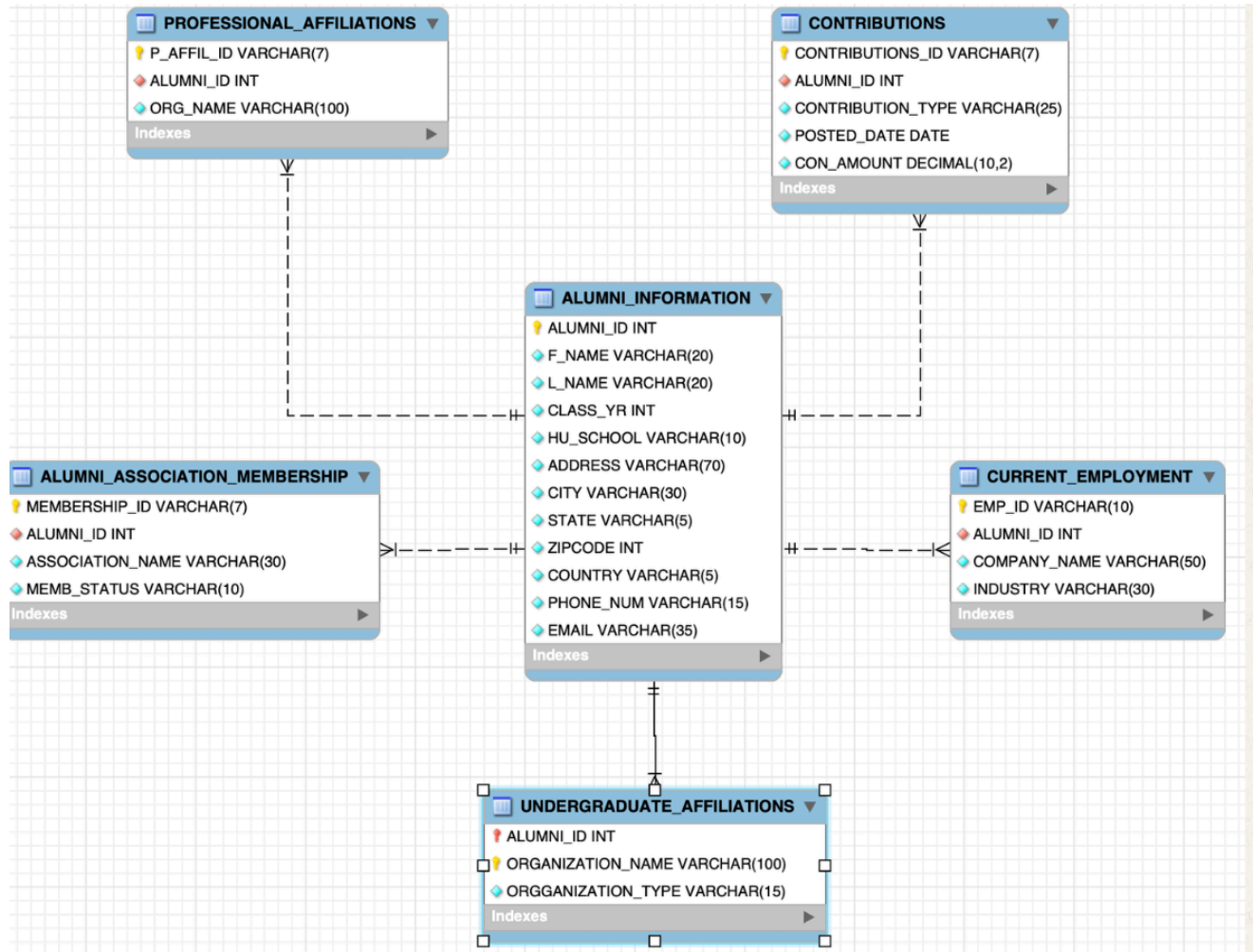
Each alumnus can have one current employment record. Each current employment record belongs to one alum.

Each alumnus can have one association membership record. Each association membership record belongs to one alum.

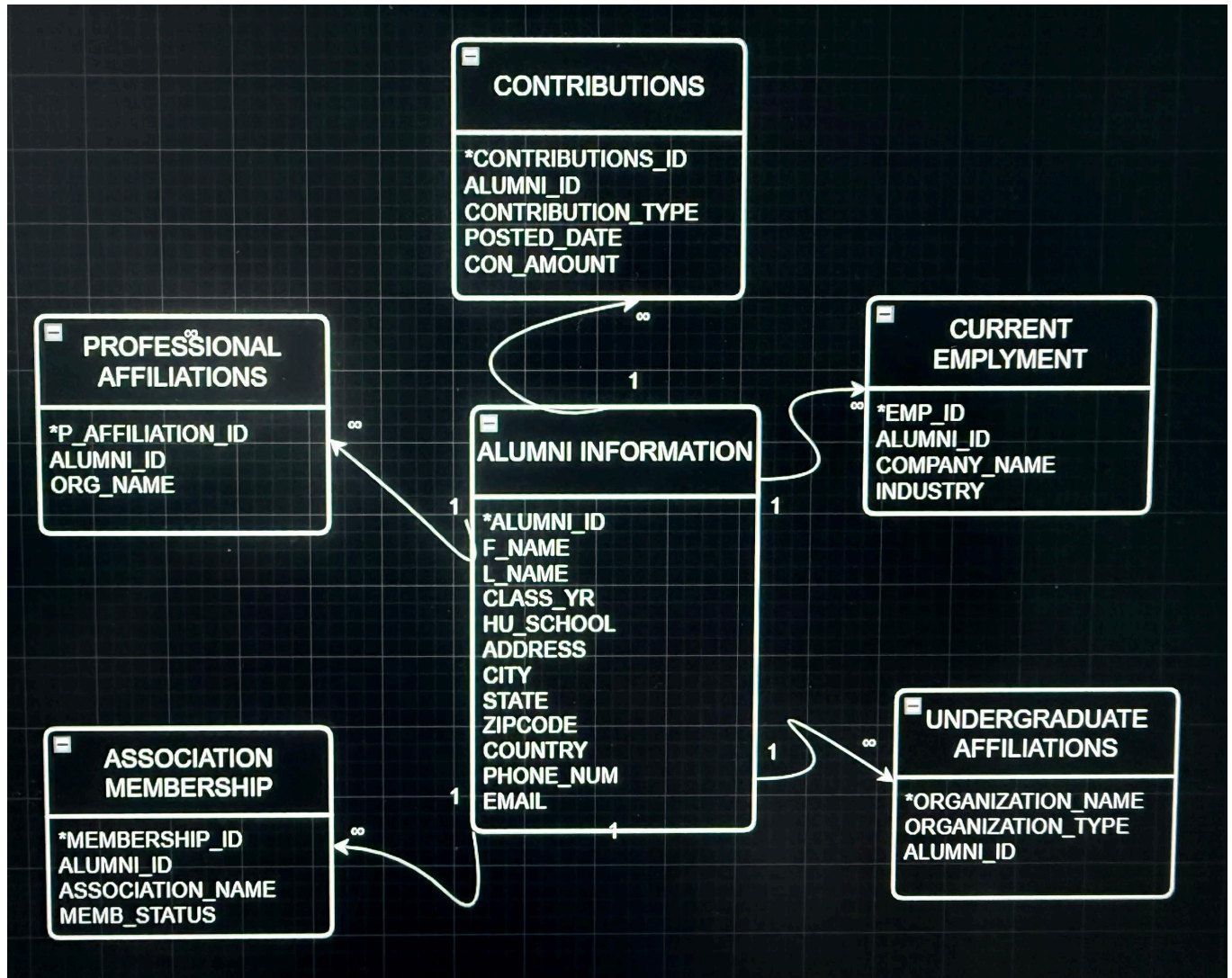
Each alumnus can have many undergraduate affiliations. Each undergraduate affiliation is associated with one alum.

Each alumnus can have many professional affiliations. Each professional affiliation is associated with one alum.

Entity Relationship Diagram:



Relational Diagram:



Normalization:

The Alumni Sub-System tables are fully normalized to Third Normal Form (3NF). This helps keep data accurate, removes redundancy, and makes database operations more efficient.

Each table avoids repeated groups and uses a unique primary key for each record.

Each table uses a single-attribute primary key, and there are no partial dependencies. All non-key attributes depend only on their table's primary key. For example, EMP_ID determines employment details, and CONTRIBUTION_ID determines contribution information. This setup avoids duplication and makes sure each table covers just one clear topic.

The subsystem also contains no transitive dependencies. No non-key attribute depends on another non-key attribute in any table. For example, in ALUMNI_INFORMATION, fields such as CITY, STATE, EMAIL, and PHONE depend only on ALUMNI_ID, not on each other. The same is true for tables like CURRENT_EMPLOYMENT, PROFESSIONAL_AFFILIATION, and ALUMNI_ASSOCIATION_MEMBERSHIP, where all details depend only on their primary keys. This design keeps each table focused and helps prevent problems when updating, adding, or deleting data.

By being normalized in 3NF, the Alumni Sub-System maintains a well-organized structure, supports accurate reporting, reduces redundancy, and ensures data remains consistent and reliable across all six entities.

Data Dictionaries:

ALUMNI_INFORMATION Table				
Column Name	Data Type	Field Length	Constraint	Description
<u>ALUMNI_ID</u>	Int	N/A	Not Null, Primary Key	Alum's ID, auto-generated
F_NAME	Varchar	20	Not Null	First name of alum
L_NAME	Varchar	20	Not Null	Last name of alum
CLASS_YR	Int	N/A	Not Null	Alum's graduation year
HU_SCHOOL	Varchar	40	Not Null	Alum Howard School
ADDRESS	Varchar	35	Not Null	Alum's address
CITY	Varchar	20	Not Null	Alum's city
STATE	Varchar	35	Not Null	Alum's state
ZIPCODE	Varchar	30	Not Null	Alum's zipcode
COUNTRY	Varchar	10	Not Null	Alum's country
C_PHONE	Varchar	15	Not Null	Alum's phone no.
C_EMAIL	Varchar	30	Not Null	Alum's email

CONTRIBUTIONS Table				
Column Name	Data Type	Field Length	Constraint	Description
<u>CONTRIBUTION_ID</u>	Varchar	7	Not Null, Primary Key	Alum's contribution ID, auto-generated
ALUMNI_ID	Int	N/A	Not Null, Foreign Key	Alum's ID, auto-generated
CON_TYPE	Varchar	15		Alum's contribution type, credit/debit
POSTED_DATE	Date	N/A		Date Alum's contribution posted
CON_AMT	Decimal	(10,2)		Alum's contribution amount

CURRENT EMPLOYMENT Table				
Column Name	Data Type	Field Length	Constraint	Description
EMP_ID	Varchar	10	Not Null, Primary Key	Company ID, auto-generated
ALUMNI_ID	Int	N/A	Not Null, Foreign Key	Alum ID, auto-generated
CO_NAME	Varchar	50		Name of the company
INDUSTY	Varchar	30		Industry of the company

UNDERGRADUATE_AFFILIATION Table				
Column Name	Data Type	Field Length	Constraint	Description
ORG_NAME	Varchar	100	Not Null, Primary Key	Name of organization
ORG_TYPE	Varchar	15		Organization type
ALUMNI_ID	Int	N/A	Not Null, Foreign Key	Alum ID, auto-generated

PROFESSIONAL_AFFILIATION Table				
Column Name	Data Type	Field Length	Constraint	Description
P_AFFL_ID	Varchar	7	Not Null, Primary Key	Professional Affiliation ID, auto-generated
ALUMNI_ID	Int	N/A	Not Null, Foreign Key	Alum ID, auto-generated
ORG_NAME	Varchar	100		Name of organization

ALUMNI_ASSOCIATION_MEMBERSHIP Table				
Column Name	Data Type	Field Length	Constraint	Description
MEMBERSHIP_ID	Varchar	7	Not Null, Primary Key	Membership ID, auto-generated
ALUMNI_ID	Int	N/A	Not Null, Foreign Key	Alum ID, auto-generated
ASSOCIATION_NAME	Varchar	30		Association name
MEMB_STATUS	Varchar	10		Alum membership status, active or inactive

SQL:

```

1 • DROP DATABASE Alumni_DMProject;
2 • CREATE DATABASE Alumni_DMProject;
3 • USE Alumni_DMProject;
4  /*Alumni Subsystem Database Design
5   Project members: Brooklyn, Lecia, Naimah, Sophia, Bridgette
6   Dr. Owunnwane Spring 2026 */
7
8   /* ALUMNI_INFORMATION TABLE */
9 • CREATE TABLE ALUMNI_INFORMATION
10  (ALUMNI_ID INT NOT NULL,
11   F_NAME VARCHAR(20) NOT NULL,
12   L_NAME VARCHAR(20) NOT NULL,
13   CLASS_YR INT NOT NULL,
14   HU_SCHOOL VARCHAR(40) NOT NULL,
15   ADDRESS VARCHAR(35) NOT NULL,
16   CITY VARCHAR(20) NOT NULL,
17   STATE VARCHAR(35) NOT NULL,
18   ZIPCODE VARCHAR(30) NOT NULL,
19   COUNTRY VARCHAR(10) NOT NULL,
20   C_PHONE VARCHAR(15) NOT NULL,
21   C_EMAIL VARCHAR(30) NOT NULL,
22   PRIMARY KEY (ALUMNI_ID));

```



```
/* CONTRIBUTIONS TABLE */
CREATE TABLE CONTRIBUTIONS
(CONTRIBUTION_ID VARCHAR(7) NOT NULL,
ALUMNI_ID INT NOT NULL,
CON_TYPE VARCHAR(15),
POSTED_DATE DATE,
CON_AMT DECIMAL(10,2),
PRIMARY KEY (CONTRIBUTION_ID),
FOREIGN KEY (ALUMNI_ID) REFERENCES ALUMNI_INFORMATION(ALUMNI_ID));

INSERT INTO CONTRIBUTIONS VALUES
("C01", 1, "Card", "2025-11-15", 250.00),
("C02", 2, "Cash", "2024-05-20", 50.00),
("C03", 3, "Card", "2026-01-10", 5000.00),
("C04", 4, "Card", "2024-09-12", 100.00),
("C05", 5, NULL, NULL, 0.00),
("C06", 6, "Card", "2024-03-12", 100.00),
("C07", 7, "Cash", "2023-10-01", 20.00),
("C08", 8, NULL, NULL, 0.00),
("C09", 9, "Card", "2022-12-05", 300.00),
("C10", 10, "Card", "2025-02-14", 75.00);
SELECT * FROM CONTRIBUTIONS;
```


Result Grid					
Filter Rows:					
Edit:					
Export					
	CONTRIBUTION_ID	ALUMNI_ID	CON_TYPE	POSTED_DATE	CON_AMT
▶	C01	1	Card	2025-11-15	250.00
	C02	2	Cash	2024-05-20	50.00
	C03	3	Card	2026-01-10	5000.00
	C04	4	Card	2024-09-12	100.00
	C05	5	NULL	NULL	0.00
	C06	6	Card	2024-03-12	100.00
	C07	7	Cash	2023-10-01	20.00
	C08	8	NULL	NULL	0.00
	C09	9	Card	2022-12-05	300.00
	C10	10	Card	2025-02-14	75.00
✱	NULL	NULL	NULL	NULL	NULL

```
62
63      /* CURRENT EMPLOYMENT TABLE */
64 • CREATE TABLE CURRENT_EMPLOYMENT
65   (EMP_ID VARCHAR(10) NOT NULL,
66    ALUMNI_ID INT NOT NULL,
67    CO_NAME VARCHAR(50),
68    INDUSTRY VARCHAR(30),
69    PRIMARY KEY (EMP_ID),
70    FOREIGN KEY (ALUMNI_ID) REFERENCES ALUMNI_INFORMATION(ALUMNI_ID));
71
72 • INSERT INTO CURRENT_EMPLOYMENT VALUES
73   ("EMP01", 1, "Microsoft", "Technology"),
74   ("EMP02", 2, "Delta Air Lines", "Aviation"),
75   ("EMP03", 3, "Dept. of Education", "Government"),
76   ("EMP04", 4, "Howard University", "Higher Education"),
77   ("EMP05", 5, NULL, NULL),
78   ("EMP06", 6, "Amazon", "Technology"),
79   ("EMP07", 7, "Bank of America", "Finance"),
80   ("EMP08", 8, "CNN", "Media"),
81   ("EMP09", 9, "Booz Allen Hamilton", "Consulting"),
82   ("EMP10", 10, "Kaiser Permanente", "Healthcare");
83 • SELECT * FROM CURRENT_EMPLOYMENT;
```

Result Grid					Filter Rows:	Edit:
	EMP_ID	ALUMNI_ID	CO_NAME	INDUSTY		
▶	EMP01	1	Microsoft	Technology		
	EMP02	2	Delta Air Lines	Aviation		
	EMP03	3	Dept. of Education	Government		
	EMP04	4	Howard University	Higher Education		
	EMP05	5	NULL	NULL		
	EMP06	6	Amazon	Technology		
	EMP07	7	Bank of America	Finance		
	EMP08	8	CNN	Media		
	EMP09	9	Booz Allen Hamilton	Consulting		
	EMP10	10	Kaiser Permanente	Healthcare		
●	NULL	NULL	NULL	NULL		

```

86      /* UNDERGRADUATE_AFFILIATION TABLE */
87 •   CREATE TABLE UNDERGRADUATE_AFFILIATION
88     (ORG_NAME VARCHAR(100) NOT NULL,
89      ORG_TYPE VARCHAR(15),
90      ALUMNI_ID INT NOT NULL,
91      PRIMARY KEY (ORG_NAME),
92      FOREIGN KEY (ALUMNI_ID) REFERENCES ALUMNI_INFORMATION(ALUMNI_ID));
93
94 •   INSERT INTO UNDERGRADUATE_AFFILIATION VALUES
95     ("Alpha Kappa Alpha Sorority, Inc.", "Sorority", 1),
96     ("Delta Sigma Theta Sorority, Inc.", "Sorority", 2),
97     ("Phi Beta Sigma Fraternity, Inc.", "Fraternity", 3),
98     ("N/A", NULL, 4),
99     ("Howard University Student Association", "Academic", 5),
100    ("National Society of Black Engineers - HU", "Professional", 6),
101    ("African Students Association - Howard", "Cultural", 7),
102    ("The Hilltop Newspaper", "Media", 8),
103    ("Howard University Alternative Spring Break", "Service", 9),
104    ("Executive Leadership Honors Program", "Professional", 10);
105 •   SELECT * FROM UNDERGRADUATE_AFFILIATION Order by ALUMNI_ID;

```

Result Grid			
Filter Rows:			
Edit:			
	ORG_NAME	ORG_TYPE	ALUMNI_ID
▶	Alpha Kappa Alpha Sorority, Inc.	Sorority	1
	Delta Sigma Theta Sorority, Inc.	Sorority	2
	Phi Beta Sigma Fraternity, Inc.	Fraternity	3
	N/A	NULL	4
	Howard University Student Association	Academic	5
	National Society of Black Engineers – HU	Professional	6
	African Students Association – Howard	Cultural	7
	The Hilltop Newspaper	Media	8
	Howard University Alternative Spring Break	Service	9
	Executive Leadership Honors Program	Professional	10
✱	NULL	NULL	NULL

```

107      /* PROFESSIONAL_AFFILIATION TABLE */
108 • CREATE TABLE PROFESSIONAL_AFFILIATION
109   (P_AFFL_ID VARCHAR(7) NOT NULL,
110    ALUMNI_ID INT NOT NULL,
111    ORG_NAME VARCHAR(100),
112    PRIMARY KEY (P_AFFL_ID),
113    FOREIGN KEY (ALUMNI_ID) REFERENCES ALUMNI_INFORMATION(ALUMNI_ID));
114
115 • INSERT INTO PROFESSIONAL_AFFILIATION VALUES
116   ("PA01", 1, "ACM"),
117   ("PA02", 2, "American Marketing Association"),
118   ("PA03", 3, "APSIA"),
119   ("PA04", 4, "African Christian Fellows"),
120   ("PA05", 5, "CFA Institute"),
121   ("PA06", 6, "IEEE"),
122   ("PA07", 7, "NABA"),
123   ("PA08", 8, "NABJ"),
124   ("PA09", 9, "AMS"),
125   ("PA10", 10, "APA");
126 • SELECT * FROM PROFESSIONAL_AFFILIATION;

```

Result Grid			
Filter Rows:			
Edit:			
	P_AFFL_ID	ALUMNI_ID	ORG_NAME
▶	PA01	1	ACM
	PA02	2	American Marketing Association
	PA03	3	APSIA
	PA04	4	African Christian Fellows
	PA05	5	CFA Institute
	PA06	6	IEEE
	PA07	7	NABA
	PA08	8	NABJ
	PA09	9	AMS
	PA10	10	APA
⬇	NULL	NULL	NULL

```

129      /* ALUMNI_ASSOCIATION_MEMBERSHIP TABLE */
130  • CREATE TABLE ALUMNI_ASSOCIATION_MEMBERSHIP
131      (MEMERSHIP_ID VARCHAR(7) NOT NULL,
132       ALUMNI_ID INT NOT NULL,
133       ASSOCIATION_NAME VARCHAR(30),
134       MEMB_STATUS VARCHAR(10),
135       PRIMARY KEY (MEMERSHIP_ID),
136       FOREIGN KEY (ALUMNI_ID) REFERENCES ALUMNI_INFORMATION(ALUMNI_ID));
137
138  • INSERT INTO ALUMNI_ASSOCIATION_MEMBERSHIP VALUES
139      ("M01", 1, "HU Alumni Assoc.", "Active"),
140      ("M02", 2, "HU Alumni Assoc.", "Active"),
141      ("M03", 3, "HU Alumni Assoc.", "Inactive"),
142      ("M04", 4, NULL, NULL),
143      ("M05", 5, "HU Alumni Assoc.", "Inactive"),
144      ("M06", 6, "HU Alumni Assoc.", "Active"),
145      ("M07", 7, "HU Alumni Assoc.", "Active"),
146      ("M08", 8, "HU Alumni Assoc.", "Inactive"),
147      ("M09", 9, "HU Alumni Assoc.", "Active"),
148      ("M10", 10, "HU Alumni Assoc.", "Active");
149  • SELECT * FROM ALUMNI_ASSOCIATION_MEMBERSHIP;

```


	MEMERSHIP_ID	ALUMNI_ID	ASSOCIATION_NAME	MEMB_STATUS
▶	M01	1	HU Alumni Assoc.	Active
	M02	2	HU Alumni Assoc.	Active
	M03	3	HU Alumni Assoc.	Inactive
	M04	4	NULL	NULL
	M05	5	HU Alumni Assoc.	Inactive
	M06	6	HU Alumni Assoc.	Active
	M07	7	HU Alumni Assoc.	Active
	M08	8	HU Alumni Assoc.	Inactive
	M09	9	HU Alumni Assoc.	Active
	M10	10	HU Alumni Assoc.	Active
✱	NULL	NULL	NULL	NULL

GUI Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<title>howard-alumni-subsystem-final</title>
```

```
<link rel="preconnect" href="https://fonts.googleapis.com">
```

```
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
```

```
<link
```

```
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
```

```
<style>
```

```
* { box-sizing: border-box; }
```

```
body { margin: 0; font-family: 'Inter', sans-serif; background: #f5f3ee; color: #1c2430; }
```

```
img { display: block; max-width: 100%; }
```

```
button, input, select { font: inherit; }
```

```
a { text-decoration: none; }
```

```
.hidden { display: none !important; }
```

```
.login-screen { min-height: 100vh; display: grid; grid-template-columns: 1.25fr .95fr; }
```

```
.login-left {
```

```
  background: #0b4670;
```

```
  color: white;
```

```
  padding: 58px;
```

```
  display: flex;
```

```
  align-items: flex-start;
```

```
  position: relative;
```

```
  overflow: hidden;
```

```
}
```

```
.login-left::before {
```

```
  content: ";
```

```
  position: absolute;
```

```
  width: 420px;
```

```
height: 420px;

border: 1px solid rgba(255,255,255,.08);

border-radius: 50%;

right: 10%;

top: 14%;

}

.login-left::after {

content: "";

position: absolute;

width: 560px;

height: 1px;

background: rgba(255,255,255,.08);

right: 0;

top: 46%;

}

.login-left-content { max-width: 560px; position: relative; z-index: 1; }

.brand-chip { display: inline-flex; align-items: center; gap: 12px; margin-bottom: 34px; }
```

```
.brand-chip img { width: 42px; height: 42px; object-fit: contain; border-radius: 9px;
background: rgba(255,255,255,.11); padding: 4px; }

.brand-chip span { font-weight: 600; }

.login-left h1 { margin: 0 0 18px; font-size: 56px; line-height: 1.05; max-width: 8.5ch; }

.login-left h2 { margin: 0 0 18px; font-size: 24px; font-weight: 500; color:
rgba(255,255,255,.92); }

.login-left p { margin: 0; line-height: 1.6; color: rgba(255,255,255,.82); max-width: 44ch; }

.login-left .office { margin-top: 26px; font-size: 14px; color: rgba(255,255,255,.66); }

.login-right { display: flex; align-items: center; justify-content: center; padding: 40px;
background: #f5f3ee; }

.login-card {

width: min(100%, 420px);

background: white;

border: 1px solid #e4dfd8;

border-radius: 18px;

box-shadow: 0 18px 40px rgba(0,0,0,.10);
```

```
padding: 34px;

}

.login-card .mini-logo { width: 30px; height: 30px; object-fit: contain; margin-bottom: 18px; }

.login-card h3 { margin: 0 0 8px; font-size: 34px; color: #0b4670; }

.field { margin-bottom: 18px; }

.field label { display: block; margin-bottom: 8px; font-size: 13px; font-weight: 700; color:
#35506a; }

.field input, .field select {

width: 100%;

padding: 14px 14px;

border: 1px solid #d7dbe3;

border-radius: 9px;

background: #f7f8fb;

outline: none;

}

.field input:focus, .field select:focus { border-color: #0b4e78; background: white; }

.login-submit, .secondary-btn {
```

```
width: 100%;

padding: 14px 16px;

border: none;

border-radius: 9px;

background: #0b4e78;

color: white;

font-weight: 700;

cursor: pointer;

}

.message { min-height: 20px; margin-top: 14px; font-size: 14px; color: #b42318; }

.dashboard-screen { display: none; min-height: 100vh; }

.dashboard-screen.active { display: grid; grid-template-columns: 250px 1fr; }

.sidebar {

background: #0b4670;

color: white;

padding: 22px 16px;
```

```
min-height: 100vh;

display: flex;

flex-direction: column;

justify-content: space-between;

}

.sidebar-top .brand-box {

display: flex;

align-items: center;

gap: 12px;

padding: 14px;

border-bottom: 1px solid rgba(255,255,255,.10);

margin-bottom: 18px;

}

.sidebar-top .brand-box img { width: 34px; height: 34px; object-fit: contain; background:
rgba(255,255,255,.12); border-radius: 8px; padding: 3px; }

.sidebar-top .brand-box strong { display: block; }

.sidebar-top .brand-box span { font-size: 13px; color: rgba(255,255,255,.70); }
```



```
.sidebar-nav button {  
  
    width: 100%;  
  
    background: transparent;  
  
    border: 1px solid transparent;  
  
    color: white;  
  
    text-align: left;  
  
    padding: 12px 14px;  
  
    border-radius: 10px;  
  
    margin-bottom: 10px;  
  
    cursor: pointer;  
  
    font-weight: 500;  
  
}  
  
.sidebar-nav button.active, .sidebar-nav button:hover {  
  
    background: rgba(255,255,255,.12);  
  
    border-color: rgba(255,255,255,.08);  
  
}  
  
.sidebar-footer {
```

```
border-top: 1px solid rgba(255,255,255,.10);  
  
padding: 16px 14px 0;  
  
}
```

```
.sidebar-footer .user-box {  
  
background: rgba(255,255,255,.08);  
  
border-radius: 10px;  
  
padding: 12px;  
  
margin-bottom: 12px;  
  
font-size: 14px;  
  
}
```

```
.sidebar-footer .signout {  
  
width: 100%;  
  
background: transparent;  
  
border: none;  
  
color: white;  
  
text-align: left;  
  
padding: 10px 8px;
```

```
cursor: pointer;
```

```
font-size: 15px;
```

```
}
```

```
.main-area { padding: 28px; }
```

```
.page-head { margin-bottom: 20px; }
```

```
.page-head h1 { margin: 0 0 6px; font-size: 28px; color: #0b4670; }
```

```
.page { display: none; }
```

```
.page.active { display: block; }
```

```
.hero-banner {
```

```
    background: linear-gradient(rgba(9,49,82,.84), rgba(9,49,82,.84)), url('howard.jpg')
```

```
center/cover no-repeat;
```

```
border-radius: 18px;
```

```
padding: 26px;
```

```
color: white;
```

```
display: flex;
```

```
justify-content: space-between;

gap: 20px;

align-items: center;

margin-bottom: 20px;

}

.hero-banner h2 { margin: 0 0 8px; font-size: 30px; }

.hero-banner .hero-actions { display: flex; gap: 10px; flex-wrap: wrap; }

.secondary-btn {

width: auto;

background: white;

color: #0b4e78;

border: 1px solid #c7d2de;

}

.stats { display: grid; grid-template-columns: repeat(4, 1fr); gap: 16px; margin-bottom: 20px;

}

.stat-card, .panel {
```

```
background: white;

border: 1px solid #e5dfd8;

border-radius: 16px;

box-shadow: 0 14px 30px rgba(0,0,0,.06);

}

.stat-card { padding: 20px; }

.stat-card span { display: block; font-size: 12px; text-transform: uppercase; letter-spacing:
.08em; color: #79808a; margin-bottom: 10px; }

.stat-card strong { font-size: 30px; color: #0b4670; }

.grid-two { display: grid; grid-template-columns: 1fr 1fr; gap: 18px; }

.panel { padding: 22px; }

table { width: 100%; border-collapse: collapse; }

th, td { text-align: left; padding: 12px 8px; border-bottom: 1px solid #eee8e1; font-size: 14px;
}

th { font-size: 12px; color: #79808a; text-transform: uppercase; letter-spacing: .06em; }
```

```
.profile-grid { display: grid; grid-template-columns: .92fr 1.08fr; gap: 18px; }
```

```
.profile-card-main { text-align: center; }
```

```
.profile-avatar-wrap {
```

```
  width: 150px;
```

```
  height: 150px;
```

```
  border-radius: 999px;
```

```
  margin: 0 auto 16px;
```

```
  border: 1px solid #e5dfd8;
```

```
  display: flex;
```

```
  align-items: center;
```

```
  justify-content: center;
```

```
  background: #fff;
```

```
  overflow: hidden;
```

```
}
```

```
.profile-avatar-wrap img { width: 120px; height: 120px; object-fit: contain; }
```

```
.profile-card-main h2 { margin: 0 0 6px; color: #0b4670; font-size: 24px; }
```

```
.profile-id { color: #c34666; font-weight: 700; margin-bottom: 10px; }
```

```
.status-pill { display: inline-block; padding: 7px 14px; border-radius: 999px; background:
#0b4e78; color: white; font-size: 13px; margin-bottom: 22px; }
```

```
.contact-box { border-top: 1px solid #ece6df; padding-top: 18px; text-align: left; }
```

```
.contact-box h4 { margin: 0 0 12px; color: #0b4670; font-size: 20px; }
```

```
.contact-box p { margin: 8px 0; color: #5f6976; }
```

```
.info-card .info-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 22px; }
```

```
.info-item small { display: block; color: #79808a; margin-bottom: 6px; }
```

```
.info-item strong { color: #0b4670; font-size: 18px; }
```

```
.donation-total { float: right; text-align: right; color: #d43b5f; font-weight: 800; font-size:
18px; }
```

```
.donation-row { display: flex; justify-content: space-between; align-items: center; padding:
16px 12px; border: 1px solid #eee8e1; border-radius: 12px; margin-bottom: 12px; }
```

```
.donation-row:last-child { margin-bottom: 0; }
```

```
.donation-row strong { color: #0b4670; }
```

```
.donation-row span { display: block; color: #7a808c; font-size: 13px; }
```

```
.donation-row .amount { color: #d43b5f; font-size: 18px; font-weight: 800; }
```

```
.bullet-list { margin: 0; padding-left: 18px; color: #5f6976; }
```

```
.bullet-list li { margin-bottom: 10px; }
```

```
@media (max-width: 1100px) {
```

```
  .dashboard-screen.active, .login-screen, .grid-two, .profile-grid, .info-card .info-grid, .stats {  
    grid-template-columns: 1fr; }
```

```
  .dashboard-screen.active { display: block; }
```

```
  .sidebar { min-height: auto; }
```

```
  .hero-banner { flex-direction: column; align-items: flex-start; }
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<section id="loginPage" class="login-screen">
```

```
<div class="login-left">
```

```
<div class="login-left-content">
```

```
<div class="brand-chip">
```



```


<span>Howard University</span>

</div>

<h1>Howard University Alumni Subsystem</h1>

<h2>Secure access for authorized staff</h2>

<div class="office">Office of Alumni Relations & University Administration</div>

</div>

</div>

<div class="login-right">

  <div class="login-card">

    <h3>Login</h3>

    <form id="loginForm">

      <div class="field">

        <label for="userid">USERID</label>

        <input id="userid" type="text" placeholder="Enter your user ID" required>
```

```
</div>
```

```
<div class="field">
```

```
<label for="password">Password</label>
```

```
<input id="password" type="password" placeholder="Enter your password" required>
```

```
</div>
```

```
<button class="login-submit" type="submit">Log In</button>
```

```
<p id="loginMessage" class="message"></p>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<section id="dashboardPage" class="dashboard-screen">
```

```
<aside class="sidebar">
```

```
<div class="sidebar-top">
```

```
<div class="brand-box">
```

```

```

<div>

Howard University

Alumni Subsystem

</div>

</div>

<nav class="sidebar-nav">

<button class="nav-btn active" data-page="dashboard">Dashboard</button>

<button class="nav-btn" data-page="directory">Alumni Directory</button>

<button class="nav-btn" data-page="contributions">Contributions</button>

<button class="nav-btn" data-page="profile">Alumni Profile</button>

<button class="nav-btn" data-page="employment">Employment</button>

</nav>

</div>

<div class="sidebar-footer">

<div class="user-box">

Staff User

staff@howard.edu

```
</div>
```

```
<button id="signOutBtn" class="signout">Sign Out</button>
```

```
</div>
```

```
</aside>
```

```
<main class="main-area">
```

```
<div class="page-head">
```

```
<h1 id="pageTitle">Dashboard</h1>
```

```
</div>
```

```
<section id="dashboardSection" class="page active">
```

```
<div class="hero-banner">
```

```
<div>
```

```
<h2>Howard University Alumni Subsystem</h2>
```

```
</div>
```

```
<div class="hero-actions">
```

```
<button class="secondary-btn" data-jump="contributions">New Contribution</button>
```

```
<button class="secondary-btn" data-jump="profile">Open Profile</button>
```

```
</div>
```

```
</div>
```

```
<div class="stats">
```

```
<div class="stat-card"><span>Alumni Records</span><strong>18,452</strong></div>
```

```
<div class="stat-card"><span>Contribution
```

```
Total</span><strong>$1,000,000</strong></div>
```

```
<div class="stat-card"><span>Memberships</span><strong>6,984</strong></div>
```

```
<div class="stat-card"><span>Employment
```

```
Records</span><strong>11,206</strong></div>
```

```
</div>
```

```
<div class="grid-two">
```

```
<div class="panel">
```

```
<h3>Alumni Directory</h3>
```

```
<table>
```

```
<thead>
```

```

<tr><th>Alumni ID</th><th>First Name</th><th>Last Name</th><th>Class
Year</th><th>Email</th><th>School</th></tr>

</thead>

<tbody>

<tr><td>1</td><td>Adam</td><td>Mathews</td><td>2023</td><td>adam.mathews@gmail.co
m</td><td>COAS</td></tr>

<tr><td>2</td><td>Navaeh</td><td>Thompson</td><td>2024</td><td>nthompson@gmail.co
m</td><td>School of Business</td></tr>

<tr><td>3</td><td>Jayden</td><td>Belle</td><td>2018</td><td>jayden.belle@gmail.com</t
d><td>COAS</td></tr>

<tr><td>4</td><td>Daniel</td><td>Owunwanne</td><td>2000</td><td>daniel.owunwanne@
howard.edu</td><td>College of Engineering and Architecture</td></tr>

<tr><td>5</td><td>Darrian</td><td>Coleman</td><td>2020</td><td>dcoleman@gmail.com<
/td><td>School of Business</td></tr>

```

```
<tr><td>6</td><td>Aaliyah</td><td>Green</td><td>2022</td><td>agreen@gmail.com</td><td>School of Business</td></tr>
```

```
<tr><td>7</td><td>Marcus</td><td>Wright</td><td>2021</td><td>mwright@gmail.com</td><td>COAS</td></tr>
```

```
<tr><td>8</td><td>Sofia</td><td>Lopez</td><td>2025</td><td>slopez@gmail.com</td><td>SOC</td></tr>
```

```
<tr><td>9</td><td>Ethan</td><td>Johnson</td><td>2019</td><td>ejohnson@gmail.com</td><td>COAS</td></tr>
```

```
<tr><td>10</td><td>Brianna</td><td>Davis</td><td>2024</td><td>bdavis@gmail.com</td><td>COAS</td></tr>
```

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<section id="directorySection" class="page">
```

```
<div class="panel">
```

```
<h3>Alumni Directory</h3>
```

```
<table>
```

```
<thead>
```

```
<tr><th>Alumni ID</th><th>First Name</th><th>Last Name</th><th>Class  
Year</th><th>Email</th><th>School</th></tr>
```

```
</thead>
```

```
<tbody>
```

```
<tr><td>1</td><td>Adam</td><td>Mathews</td><td>2023</td><td>adam.mathews@gmail.co  
m</td><td>COAS</td></tr>
```

```
<tr><td>2</td><td>Navaeh</td><td>Thompson</td><td>2024</td><td>nthompson@gmail.co  
m</td><td>School of Business</td></tr>
```

```
<tr><td>3</td><td>Jayden</td><td>Belle</td><td>2018</td><td>jayden.belle@gmail.com</t  
d><td>COAS</td></tr>
```


4	Daniel	Owunwanne	2000	daniel.owunwanne@howard.edu	College of Engineering and Architecture
---	--------	-----------	------	-----------------------------	---

5	Darrian	Coleman	2020	dcoleman@gmail.com	School of Business
---	---------	---------	------	--------------------	--------------------

6	Aaliyah	Green	2022	agreen@gmail.com	School of Business
---	---------	-------	------	------------------	--------------------

7	Marcus	Wright	2021	mwright@gmail.com	COAS
---	--------	--------	------	-------------------	------

8	Sofia	Lopez	2025	slopez@gmail.com	SOC
---	-------	-------	------	------------------	-----

9	Ethan	Johnson	2019	ejohnson@gmail.com	COAS
---	-------	---------	------	--------------------	------

10	Brianna	Davis	2024	bdavis@gmail.com	COAS
----	---------	-------	------	------------------	------

```
</tbody>
```

```
</table>
```

```
</div>
```

```
</section>
```

```
<section id="contributionsSection" class="page">
```

```
<div class="grid-two">
```

```
<div class="panel">
```

```
<h3>Record Contribution</h3>
```

```
<button id="openPaymentModal" class="secondary-btn" type="button"
style="margin-bottom: 16px;">Put Card Info Here</button>
```

```
<form id="contributionForm">
```

```
<div class="field">
```

```
<label for="contribAlumniId">Alumni ID</label>
```

```
<input id="contribAlumniId" type="text" placeholder="Enter alumni ID" required>
```

```
</div>
```

```
<div class="field">
```

<label for="contribAmount">Contribution Amount</label>

<input id="contribAmount" type="number" step="0.01" placeholder="0.00" required>

</div>

<div class="field">

<label for="contribType">Contribution Type</label>

<select id="contribType">

<option>Card</option>

<option>Cash</option>

<option>Check</option>

<option>Online</option>

</select>

</div>

<div class="field">

<label for="contribAffl">Affiliation ID</label>

<input id="contribAffl" type="text" placeholder="Enter affiliation ID">

</div>

<button class="login-submit" type="submit">Save Contribution</button>

```
<p id="contributionMessage" class="message"></p>

</form>

</div>

<div class="panel">

  <h3>Contribution Records</h3>

  <table>

    <thead>

      <tr><th>Contribution ID</th><th>Alumni
ID</th><th>Type</th><th>Date</th><th>Amount</th></tr>

    </thead>

    <tbody id="contribTableBody">

      <tr><td>C01</td><td>1</td><td>Card</td><td>2025-11-15</td><td>$250.00</td></tr>

      <tr><td>C02</td><td>2</td><td>Cash</td><td>2024-05-20</td><td>$50.00</td></tr>

      <tr><td>C03</td><td>3</td><td>Card</td><td>2026-01-10</td><td>$5,000.00</td></tr>
```

```
<tr><td>C04</td><td>4</td><td>Card</td><td>2024-09-12</td><td>$100.00</td></tr>
```

```
<tr><td>C05</td><td>5</td><td>-</td><td>-</td><td>$0.00</td></tr>
```

```
<tr><td>C06</td><td>6</td><td>Card</td><td>2024-03-12</td><td>$100.00</td></tr>
```

```
<tr><td>C07</td><td>7</td><td>Cash</td><td>2023-10-01</td><td>$20.00</td></tr>
```

```
<tr><td>C08</td><td>8</td><td>-</td><td>-</td><td>$0.00</td></tr>
```

```
<tr><td>C09</td><td>9</td><td>Card</td><td>2022-12-05</td><td>$300.00</td></tr>
```

```
<tr><td>C10</td><td>10</td><td>Card</td><td>2025-02-14</td><td>$75.00</td></tr>
```

```
</tbody>
```

```
</table>
```

```
<h3 style="margin-top:22px;">Donation History <span class="donation-total"
id="totalDonation">$1,000,000</span></h3>
```

```
<div id="donationList">
```

```
<div class="donation-row">
```



```
<div class="profile-grid">
```

```
<div class="panel profile-card-main">
```

```
<div class="profile-avatar-wrap"
```

```
style="background:#0b4670;color:white;font-weight:800;font-size:34px;">DO</div>
```

```
<h2>Daniel Owunwanne</h2>
```

```
<div class="profile-id">@000400</div>
```

```
<div class="status-pill">Employed</div>
```

```
<div class="contact-box">
```

```
<h4>Contact Information</h4>
```

```
<p>daniel.owunwanne@howard.edu</p>
```

```
<p>(202) 555-0147</p>
```

```
<p>Washington, DC 20001</p>
```

```
</div>
```

```
</div>
```

```
<div style="display:grid; gap:18px;">
```

```
<div class="panel info-card">
```

<h3>Academic History</h3>

<div class="info-grid">

<div class="info-item"><small>Degree</small>Bachelor of
Science</div>

<div class="info-item"><small>Major</small>Information
Systems</div>

<div class="info-item"><small>Graduation
Year</small>2000</div>

<div class="info-item"><small>GPA</small>5.0</div>

</div>

</div>

<div class="panel info-card">

<h3>Employment Information</h3>

<div class="info-grid">

<div class="info-item"><small>Current Employer</small>Howard
University</div>


```
<div class="info-item"><small>Position</small><strong>Howard University  
Professor</strong></div>
```

```
<div class="info-item"><small>Industry</small><strong>Higher  
Education</strong></div>
```

```
<div class="info-item"><small>Employment  
Status</small><strong>Employed</strong></div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<div class="grid-two" style="margin-top:18px;">
```

```
<div class="panel">
```

```
<h3>Memberships</h3>
```

```
<ul class="bullet-list">
```

```
<li>African Christian Fellows</li>
```

```
</ul>
```

<h3 style="margin-top:28px;">Affiliations</h3>

<ul class="bullet-list">

Howard University

</div>

<div class="panel">

<h3>Donation History \$1,000,000</h3>

<div class="donation-row">

<div>Annual Fund03/15/2022</div>

<div class="amount">\$500,000</div>

</div>

<div class="donation-row">

<div>Scholarship Fund09/10/2023</div>

<div class="amount">\$250,000</div>

</div>

<div class="donation-row">

<div>Annual Fund12/01/2024</div>

```
<div class="amount">$250,000</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<section id="employmentSection" class="page">
```

```
<div class="grid-two">
```

```
<div class="panel">
```

```
<h3>Current Employment</h3>
```

```
<table>
```

```
<thead>
```

```
<tr><th>EMP ID</th><th>Alumni ID</th><th>Company  
Name</th><th>Industry</th></tr>
```

```
</thead>
```

```
<tbody>
```

```
<tr><td>EMP01</td><td>1</td><td>Microsoft</td><td>Technology</td></tr>
```

EMP02	2	Delta Air Lines	Aviation
EMP03	3	Dept. of Education	Government
EMP04	4	Howard University	Higher Education
EMP05	5	-	-
EMP06	6	Amazon	Technology
EMP07	7	Bank of America	Finance
EMP08	8	CNN	Media
EMP09	9	Booz Allen Hamilton	Consulting
EMP10	10	Kaiser Permanente	Healthcare

```
</main>
```

```
</section>
```

```
<div id="paymentModal" class="hidden"
style="position:fixed;inset:0;background:rgba(0,0,0,.55);align-items:center;justify-content:center
;z-index:9999;">
```

```
<div
style="background:white;width:min(92vw,480px);border-radius:18px;padding:26px;box-shadow
:0 24px 60px rgba(0,0,0,.25);">
```

```
<div
style="display:flex;justify-content:space-between;align-items:center;margin-bottom:18px;gap:12
px;">
```

```
<h3 style="margin:0;color:#0b4670;">Card Payment</h3>
```

```
<button id="closePaymentModal" type="button"
style="border:none;background:transparent;font-size:28px;line-height:1;cursor:pointer;">&times
</button>
```

```
</div>
```

```
<form id="paymentForm">
```

```
<div class="field"><label for="cardName">Name on Card</label><input id="cardName"
type="text" required></div>
```

```
<div class="field"><label for="cardNumber">Card Number</label><input
id="cardNumber" type="text" inputmode="numeric" placeholder="1234 5678 9012 3456"
required></div>
```

```
<div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;">
```

```
<div class="field"><label for="expDate">Expiry</label><input id="expDate" type="text"
placeholder="MM/YY" required></div>
```

```
<div class="field"><label for="cvv">CVV</label><input id="cvv" type="password"
placeholder="123" required></div>
```

```
</div>
```

```
<div class="field"><label for="paymentAmount">Amount</label><input
id="paymentAmount" type="number" step="0.01" required></div>
```

```
<button class="login-submit" type="submit">Pay Now</button>
```

```
<p id="paymentMessage" class="message"></p>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
<script>
```

```
const loginPage = document.getElementById('loginPage');
```

```
const dashboardPage = document.getElementById('dashboardPage');
```

```
const loginForm = document.getElementById('loginForm');
```

```
const loginMessage = document.getElementById('loginMessage');
```

```
const signOutBtn = document.getElementById('signOutBtn');
```

```
const navButtons = document.querySelectorAll('.nav-btn');
```

```
const pages = document.querySelectorAll('.page');
```

```
const pageTitle = document.getElementById('pageTitle');
```

```
const pageSubtitle = document.getElementById('pageSubtitle');
```

```
const contributionForm = document.getElementById('contributionForm');
```

```
const contributionMessage = document.getElementById('contributionMessage');
```

```
const jumpButtons = document.querySelectorAll('[data-jump]');
```

```
const donationList = document.getElementById('donationList');
```

```
const totalDonation = document.getElementById('totalDonation');
```

```
const openPaymentModal = document.getElementById('openPaymentModal');
```

```
const paymentModal = document.getElementById('paymentModal');

const closePaymentModal = document.getElementById('closePaymentModal');

const paymentForm = document.getElementById('paymentForm');

const paymentMessage = document.getElementById('paymentMessage');


const pageMeta = {

  dashboard: ['Dashboard', ""],

  directory: ['Alumni Directory', ""],

  contributions: ['Contributions', ""],

  profile: ['Alumni Profile', ""],

  employment: ['Employment', ""],

};


let donationTotal = 1000000;


function formatMoney(n) {
```



```

    return '$' + n.toLocaleString('en-US', { minimumFractionDigits: 0, maximumFractionDigits:
2 });

```

```

}

```

```

function goToPage(name) {

```

```

    pages.forEach(page => page.classList.remove('active'));

```

```

    document.getElementById(name + 'Section').classList.add('active');

```

```

    navButtons.forEach(btn => btn.classList.remove('active'));

```

```

    document.querySelector('[data-page="' + name + '"]').classList.add('active');

```

```

    pageTitle.textContent = pageMeta[name][0];

```

```

    pageSubtitle.textContent = pageMeta[name][1];

```

```

}

```

```

loginForm.addEventListener('submit', function(event) {

```

```

    event.preventDefault();

```

```

    const userid = document.getElementById('userid').value.trim().toLowerCase();

```

```

    const password = document.getElementById('password').value.trim().toLowerCase();

```

```
const valid = userid === 'admin' && password === 'howard123';

if (valid) {

    loginMessage.style.color = 'green';

    loginMessage.textContent = 'Login successful.';

    loginPage.classList.add('hidden');

    dashboardPage.classList.add('active');

} else {

    loginMessage.style.color = '#b42318';

    loginMessage.textContent = 'Invalid user ID or password.';

}

});

signOutBtn.addEventListener('click', function() {

    dashboardPage.classList.remove('active');

    loginPage.classList.remove('hidden');

    loginForm.reset();
```

```
loginMessage.textContent = "";

goToPage('dashboard');

window.scrollTo({ top: 0, behavior: 'smooth' });

});

navButtons.forEach(button => {

    button.addEventListener('click', function() {

        goToPage(button.dataset.page);

    });

});

jumpButtons.forEach(button => {

    button.addEventListener('click', function() {

        goToPage(button.dataset.jump);

    });

});
```

```
openPaymentModal.addEventListener('click', function() {  
  
    paymentModal.classList.remove('hidden');  
  
    paymentModal.style.display = 'flex';  
  
});
```

```
closePaymentModal.addEventListener('click', function() {  
  
    paymentModal.classList.add('hidden');  
  
    paymentModal.style.display = 'none';  
  
});
```

```
paymentModal.addEventListener('click', function(e) {  
  
    if (e.target === paymentModal) {  
  
        paymentModal.classList.add('hidden');  
  
        paymentModal.style.display = 'none';  
  
    }  
  
});
```

```

paymentForm.addEventListener('submit', function(event) {

    event.preventDefault();

    const amt = parseFloat(document.getElementById('paymentAmount').value || '0');

    if (!amt || amt <= 0) {

        paymentMessage.style.color = '#b42318';

        paymentMessage.textContent = 'Enter a valid amount.';

        return;

    }

    donationTotal += amt;

    totalDonation.textContent = formatMoney(donationTotal);

    const row = document.createElement('div');

    row.className = 'donation-row';

    row.innerHTML = `<div><strong>Card Payment</strong><span>${new
Date().toISOString().slice(0,10)}</span></div><div
class="amount">${formatMoney(amt)}</div>`;

    donationList.prepend(row);

    paymentMessage.style.color = 'green';

```

```
paymentMessage.textContent = 'Payment processed successfully.';
```

```
paymentForm.reset();
```

```
});
```

```
</script>
```

```
</body>
```

```
</html>
```

Timeline:

This development timeline is intentionally designed to move the platform from a basic tool into a fully integrated alumni ecosystem that drives long-term engagement and measurable impact. The initial 2026 launch establishes a centralized alumni database, creating a foundation for connection and data collection. Building on that, the introduction of Howard Circles and geo-based networking in 2027 is meant to increase user engagement by making the platform more community-driven and relevant to users' identities, industries, and locations. In 2028, the "Adopt-a-Bison" mentorship initiative strengthens the pipeline between students and alumni, increasing retention and fostering meaningful, structured relationships that extend beyond casual networking. The 2029 mobile app launch is a strategic move to increase accessibility and daily usage, allowing users to stay connected through real-time notifications and on-the-go interaction. Finally, the implementation of the Impact Score in 2030 introduces a system of recognition and incentives, encouraging continued participation by quantifying contributions and reinforcing a culture of giving back. Together, these developments are designed to increase engagement, retention, and economic activity within the alumni network while positioning the platform as a long-term institutional asset.

Conclusion:

The Howard University Alumni Subsystem is a dependable tool for managing key alumni information. It organizes data into six main groups: Alumni Information, Current Employment, Professional Affiliations, Undergraduate Affiliations, Alumni Association Membership, and Contributions. This structure keeps data accurate, avoids duplicates, and makes it simple to find information for both administrative and academic needs. The Entity-Relationship Diagram, relational schema, and SQL work together to support a system that maintains data quality and can grow as needed.

The team developed the subsystem after a thorough review of business rules, system needs, and data relationships. It can update alumni records while making privacy and security a top priority. By using Third Normal Form, the system keeps data consistent and lowers the risk of errors during updates or reports. The Alumni Subsystem gives a strong foundation for communication, reporting, and future links to other systems like the Transcript Subsystem. In short, this project delivers a reliable and easy-to-manage database that helps the university stay connected with and support its alumni.